Steps / what this program does :

1. **Read inputs**: takes `n`, `d`, and a PA file containing permutations (each line = one permutation in S_n).
2. **Load the PA** into memory as a list `PA`.
3. **Set radius** $r = d - 1$.
4. **Compute the identity ball once**: build $B(\mathrm{id}, r)$ exactly using **(SYT P, Q pairs → inverse RSK)** (Kong-style). Store it as `ball_id`.
5. **Repeat trials (random local search)** up to `max_trials`:
   5.1 **Pick a random center** π from `PA`.
   5.2 **Remove** π from the PA to form `PA_minus = PA \ {π}`.
   5.3 **Construct the ball around** π by **left-translation**:
   $B(\pi, r) = \{\pi \circ \tau : \tau \in B(\mathrm{id}, r)\}$.
   5.4 **Filter candidates**: from $B(\pi, r)$, keep only permutations x such that
   $d_U(x, \sigma) \geq d$ for **every** $\sigma \in \mathtt{PA\_minus}$.
   5.5 If fewer than 2 candidates, **go to next trial**.
   5.6 Otherwise, **search a pair** (c_1, c_2) among candidates with **mutual distance**
   $d_U(c_1, c_2) \geq d$.
   5.7 If no such pair, **go to next trial**.
   5.8 If found, **augment the PA**:
   $\mathtt{PA\_new} = \mathtt{PA\_minus} \cup \{c_1, c_2\}$
   (so size increases by **+1**).
   5.9 **Write outputs**: overwrite PA file with `PA_new`, save removed center and the two added permutations to `removed_center.txt` and `added_two.txt`, then stop.
6. If all trials fail, **print failure message** and do not change files.

Sample Input:

5_3_pa.txt (each line is one permutation of 1..5)

1 2 3 4 5
2 1 3 4 5
3 4 5 1 2
4 5 2 3 1
5 3 4 1 2

Sample output (typical):

PA loaded: size = 5

n=5, d=3, r=d-1=2

Building B(id, r=d-1) using RSK+InverseRSK...

|B(id, d-1)| = 78

[trial 1] removed center, candidates found = 1

[trial 2] removed center, candidates found = 3

SUCCESS on trial 2!

Removed center: 2 1 3 4 5

Add1: 2 3 1 5 4

Add2: 4 1 5 2 3

Old size=5, New size=6 (improved by +1)

Updated PA written back to: 5_3_pa.txt

Saved removed center to: removed_center.txt

Saved added two perms to: added_two.txt